

Reading Report: Can AI Agents Agree? Comparison with PBFT

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Question

Can LLM agents reliably reach consensus like distributed systems?

Focus of this report

- Compare **PBFT** and **Can AI Agents Agree?**
- Identify assumptions that no longer hold
- Explore possible research directions

PBFT vs. LLM Agent Consensus

	PBFT	Can AI Agents Agree?
Node	Deterministic process	LLM Agent
Execution	Fixed protocol	Natural language reasoning
Failure Model	Byzantine node	Byzantine + reasoning errors
Output	Identical value	Probabilistic answer
Consensus Target	State / value	Answer agreement

Key Difference

PBFT assumes correct nodes always execute the protocol faithfully, while LLM agents may misunderstand the protocol itself.

Where PBFT Assumptions Break

PBFT assumes:

- Correct nodes always follow the protocol.
- Messages are interpreted correctly.
- State transitions are deterministic.

LLM agents introduce new uncertainties:

- Hallucination
- Prompt misunderstanding
- Context forgetting
- Non-deterministic reasoning

These behaviors are not covered by the classical Byzantine fault model.

Possible Research Direction

Current paper:

Consensus $\rightarrow$ *Answer*

Possible direction:

Consensus $\rightarrow$ *Workflow State*

Examples:

- Is the research stage complete?
- Can the coding stage safely begin?
- Is the review trustworthy?

Idea:

Workflow Consensus under Semantic Byzantine Faults

Takeaways

- PBFT and LLM agents have fundamentally different assumptions.
- Correct execution can no longer be assumed for LLM agents.
- Existing work studies answer consensus.
- Workflow consensus remains largely unexplored.
- Next step:
 - Read the system model carefully.
 - Analyze the failure assumptions.
 - Compare with PBFT formally.